SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	Akash J	**Reg. No.:**	192324295
Section / Batch:	2023	**Date:**	20/10/2026

Code of Conduct

*I **Akash J / 192324295** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: **Akash J**

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Derivational Morphology and Vocabulary Expansion	Word Generation, Semantic Transformation	TP-09

Derivational Morphology and Vocabulary Expansion

1. Introduction

Derivational morphology is the study of how new words are formed by adding prefixes and suffixes to a base word, and it is one of the quickest ways to expand vocabulary. For example, the base word *happy* becomes *unhappy*, *happiness* and *happily* through derivation. Each new word has its own meaning or grammatical role, so learning a single root gives access to a whole family of related words.

This report looks at two linked ideas: **word generation**, which is building new words from a root, and **semantic transformation**, which is the change in meaning or word class that an affix brings.

2. Objectives

- Explain what derivational morphology is and how it differs from inflection.
- Generate new words from a root using common prefixes and suffixes.
- Show how affixes change a word's meaning and word class.
- Apply word families to expand everyday and academic vocabulary.

3. Word Generation

Word generation means attaching an affix to a base (root) word to form a new word. A prefix comes before the base and a suffix comes after it. Derivation creates a new word, unlike inflection (such as *walk* to *walked*), which only changes grammatical form.

- happy + un- = unhappy
- happy + -ness = happiness
- agree + dis- = disagree
- agree + -ment = agreement
- act + -ive = active
- use + re- = reuse
- use + -ful = useful
- care + -less = careless
- friend + -ship = friendship

Affixes can also be stacked to build longer words, for example *act* to *action* to *actionable* to *unactionable*.

4. Semantic Transformation

An affix can change a word's meaning, its word class (part of speech), or both. Prefixes mostly change meaning, such as negation or repetition, while suffixes mostly change word class.

- kind to unkind: opposite meaning
- write to rewrite: do again
- teach to teacher: verb becomes a noun (a person who teaches)
- beauty to beautiful: noun becomes an adjective
- quick to quickly: adjective becomes an adverb
- child to childish: noun becomes an adjective with a negative sense

The same suffix can carry different shades of meaning. For example, *-er* in *teacher* names a person, while in *printer* it names a machine. Meaning can also shift beyond the literal sum of the parts: *childish* is more negative than *childlike*, even though both come from *child*.

5. Vocabulary Expansion

Learning words as families rather than one by one multiplies vocabulary quickly. One root such as *create* gives *creation*, *creative*, *creativity*, *creator*, *recreate* and *uncreative*.

Three simple strategies help:

1. **Start from the root.** Learn the base word first, then add affixes one at a time.
2. **Learn the affix meanings.** Knowing that *un-* means not and *-less* means without lets you guess unfamiliar words such as *unfathomable* or *fearless*.
3. **Check the word class.** Use suffixes to move a word into the grammatical slot a sentence needs, for example *decide* (verb), *decision* (noun), *decisive* (adjective).

Some derivations are irregular, so a dictionary should confirm any new word before it is used in formal writing.

6. Presentation Image

Paste the screenshot or image of the presentation slide in the box below, then delete the placeholder text.

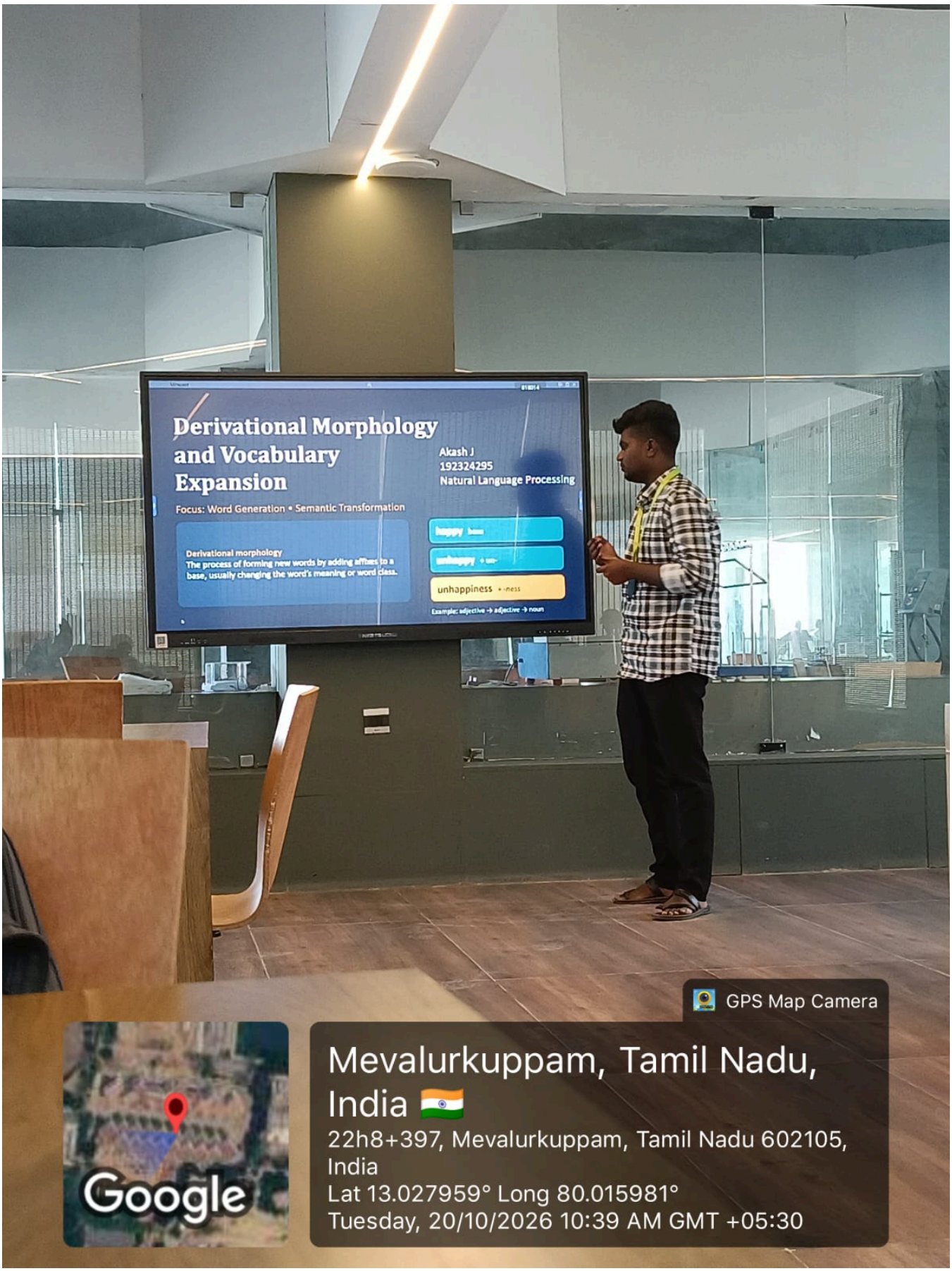


Figure 1: Slide from the presentation on Derivational Morphology and Vocabulary Expansion.

7. Conclusion

Derivational morphology gives learners a systematic way to build vocabulary. By generating new words from roots and understanding how affixes shift meaning and word class, a learner can decode unfamiliar words and use words more precisely in speech and writing.

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____